

LATACOM (LATANOPROST 0.005% + TIMOLOL 0.5%) EYE DROPS

DESCRIPTION:

Latacom (Latanoprost 0.005% + Timolol 0.5%) Eye Drops : A clear, colourless solution.

COMPOSITION:

Each 1 ml contains latanoprost 50 micrograms and timolol maleate 6.8 mg equivalent to 5 mg timolol. Contains preservative: benzalkonium chloride 0.2mg/ml

PHARMACODYNAMICS:

Pharmacotherapeutic group: Ophthalmological-beta-blocking agents - timolol, combinations

ATC code: S01ED51

Mechanism of action:

Latacom consists of two components: latanoprost and timolol maleate. These two components decrease elevated intraocular pressure (IOP) by different mechanisms of action and the combined effect results in additional IOP reduction compared to either compound administered alone.

Latanoprost, a prostaglandin F_{2α} analogue, is a selective prostanoid FP receptor agonist that reduces the IOP by increasing the outflow of aqueous humour. The main mechanism of action is increased uveoscleral outflow. Additionally, some increase in outflow facility (decrease in trabecular outflow resistance) has been reported in man. Latanoprost has no significant effect on the production of aqueous humour, the blood-aqueous barrier or the intraocular blood circulation.

Timolol is a beta-1 and beta-2 (non-selective) adrenergic receptor blocking agent that has no significant intrinsic sympathomimetic, direct myocardial depressant or membrane-stabilising activity. Timolol lowers IOP by decreasing the formation of aqueous in the ciliary epithelium.

The precise mechanism of action is not clearly established, but inhibition of the increased cyclic AMP synthesis caused by endogenous beta-adrenergic stimulation is probable. Timolol has not been found to significantly affect the permeability of the blood-aqueous barrier to plasma proteins.

PHARMACOKINETICS

Latanoprost

Absorption

Latanoprost is an isopropyl ester prodrug, which per se is inactive but after hydrolysis by esterases in the cornea to the acid of latanoprost, becomes biologically active. The prodrug is well absorbed through the cornea and all drug that enters the aqueous humor is hydrolysed during the passage through the cornea.

Distribution

Studies indicate that the maximum concentration in the aqueous humour, approximately 15-30 ng/mL, is reached about 2 hours after topical administration of latanoprost alone.

The acid of latanoprost has a plasma clearance of 0.40 l/h/kg and a small volume of distribution, 0.16 l/kg, resulting in a rapid half-life in plasma, 17 minutes. After topical ocular administration the systemic bioavailability of the acid of latanoprost is 45%. The acid of latanoprost has a plasma protein binding of 87%.

Biotransformation and elimination

There is practically no metabolism of the acid of latanoprost in the eye. The main metabolism occurs in the liver.

Timolol

Absorption and distribution

The maximum concentration of timolol in the aqueous humour is reached about 1 hour after topical administration of eye drops. Part of the dose is absorbed systemically and a maximum plasma concentration of 1 ng/mL is reached 10-20 minutes after topical administration of one eye drop to each eye once daily (300 micrograms/day).

Biotransformation

The half-life of timolol in plasma is about 6 hours. Timolol is extensively metabolised in the liver.

Elimination

The metabolites are excreted in the urine together with some unchanged timolol.

Latacom

Pharmacokinetic/pharmacodynamic relationship

No pharmacokinetic interactions between latanoprost and timolol were observed, although there was an approximate 2-fold increased concentration of the acid of latanoprost in aqueous humour 1-4 hours after administration of the combination product compared to monotherapy.

INDICATIONS:

Reduction of intraocular pressure (IOP) in patients with open-angle glaucoma and ocular hypertension who are insufficiently responsive to topical beta-blockers.

RECOMMENDED DOSAGE:

Adults (including the elderly): One drop in the affected eye(s) once daily. If one dose is missed, treatment should continue with the next dose as planned. The dose should not exceed one drop in the affected eye(s) daily. Contact lenses should be removed before instillation of the eye drops and may be reinserted after 15 minutes (See WARNINGS AND PRECAUTIONS).

If more than one topical ophthalmic drug product is being used, the drugs should be administered at least five minutes apart. When using nasolacrimal occlusion or closing the eyelids for 2 minutes, the systemic absorption is reduced. This may result in a decrease in systemic side effects and an increase in local activity.

Children and adolescents: The safety and efficacy of Latacom in children below 18 years of age has not been established.

ROUTE OF ADMINISTRATION:

Latacom must only be administered by ophthalmic route.

CONTRAINDICATIONS:

- Reactive airway disease including bronchial asthma or a history of bronchial asthma, severe chronic obstructive pulmonary disease.
- Sinus bradycardia, sick sinus syndrome, sino-atrial block, second or third degree atrioventricular block not controlled with pacemaker, overt cardiac failure, cardiogenic shock
- Hypersensitivity to the active substances or to any of the excipients

WARNINGS AND PRECAUTIONS:

This product contains benzalkonium chloride, which is commonly used as a preservative in ophthalmic products. Benzalkonium chloride has been reported to cause punctate keratopathy and/or toxic ulcerative keratopathy, may cause eye irritation and is known to discolour soft contact lenses. Close monitoring is required with frequent or prolonged use of Latacom in dry eye patients, or in conditions where the cornea is compromised. Contact lenses may absorb the benzalkonium chloride and these should be removed before applying Latacom but may be reinserted after 15 minutes.

Systemic effects

Like other topically applied ophthalmic agents, Latacom is absorbed systemically. Due to the beta-adrenergic component timolol, the same types of cardiovascular, pulmonary and other adverse reactions as seen with systemic beta-adrenergic blocking agents may occur. Incidence of systemic ADRs after topical ophthalmic administration is lower than for systemic

administration. To reduce the systemic absorption, SEE RECOMMENDED DOSAGE.

Cardiac disorders

In patients with cardiovascular diseases (e.g., coronary heart disease, Prinzmetal's Angina and cardiac failure) and hypotension therapy with beta-blockers should be critically assessed and the therapy with other active substances should be considered. Patients with cardiovascular diseases should be watched for signs of deterioration of these diseases and of adverse reactions. Due to its negative effect on conduction time, beta-blockers should only be given with caution to patients with first degree heart block. Cardiac reactions, and rarely, death in association with cardiac failures have been reported following administration of timolol.

Vascular disorders

Patients with severe peripheral circulatory disturbance/disorders (i.e. severe forms of Raynaud's disease or Raynaud's syndrome) should be treated with caution.

Respiratory disorders

Respiratory reactions, including death due to bronchospasm in patients with asthma have been reported following administration of some ophthalmic beta-blockers. Latacom should be used with caution, in patients with mild/moderate chronic obstructive pulmonary disease (COPD) and only if the potential benefit outweighs the potential risk.

Hypoglycaemia/diabetes

Beta-blockers should be administered with caution in patients subject to spontaneous hypoglycaemia or to patients with labile diabetes, as beta-blockers may mask the signs and symptoms of acute hypoglycaemia. Beta-blockers may also mask the signs of hyperthyroidism.

Corneal diseases

Ophthalmic beta-blockers may induce dryness of eyes. Patients with corneal diseases should be treated with caution.

Other beta-blocking agents

The effect on intra-ocular pressure or the known effects of systemic beta-blockade may be potentiated when timolol is given to the patients already receiving a systemic beta-blocking agent. The response of these patients should be closely observed. The use of two topical beta-adrenergic blocking agents is not recommended. (SEE INTERACTIONS WITH OTHER MEDICAMENTS)

Anaphylactic reactions

While taking beta-blockers, patients with a history of atopy or a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge with such allergens and unresponsive to the usual doses of adrenaline used to treat anaphylactic reactions.

Choroidal detachment

Choroidal detachment has been reported with administration of aqueous suppressant therapy (e.g., timolol, acetazolamide) after filtration procedures.

Surgical anesthesia

Beta-blocking ophthalmological preparations may block systemic beta-agonist effects, e.g., of adrenaline. The anaesthesiologist should be informed when the patient is receiving timolol.

Concomitant therapy

Timolol may interact with other drug. (SEE INTERACTIONS WITH OTHER MEDICAMENTS)

Other prostaglandin analogues

The concomitant use of two or more prostaglandins, prostaglandin analogues, or prostaglandin derivatives is not recommended. (SEE INTERACTIONS WITH OTHER MEDICAMENTS)

Iris pigmentation changes

Latanoprost may gradually change the eye colour by increasing the amount of brown pigment in the iris. This effect has predominantly been seen in patients with mixed coloured irides, i.e. green-brown, yellow-brown or blue/grey-brown, and is due to increased melanin content in the stromal melanocytes of the iris. Typically, the brown pigmentation around the pupil spreads concentrically towards the periphery in affected eyes, but the entire iris or parts of it may become more brownish.

The change in iris colour occurs slowly and may not be noticeable for several months to years and it has not been associated with any symptom or pathological changes.

No further increase in brown iris pigment has been observed after discontinuation of treatment, but the resultant colour change may be permanent. Neither naevi nor freckles of the iris have been affected by the treatment.

Accumulation of pigment in the trabecular meshwork or elsewhere in the anterior chamber has not been observed but patients should be examined regularly and, depending on the clinical situation, treatment may be stopped if increased iris pigmentation ensues.

Before treatment is instituted patients should be informed of the possibility of a change in eye colour. Unilateral treatment can result in permanent heterochromia.

Glaucoma

There is no documented experience with Latanoprost in inflammatory, neovascular or chronic angle closure glaucoma, in open angle glaucoma of pseudophakic patients and in pigmentary glaucoma. Latanoprost has no or little effect on the pupil but there is not documented experience in acute attacks of closed angle glaucoma. Therefore, it is recommended that Latacom should be used with caution in these conditions until more experience is obtained.

Eyelid and eyelash changes

Eyelid skin darkening, which may be reversible, has been reported in association with the use of latanoprost.

Latanoprost may gradually change eyelashes and vellus hair in the treated eye; these changes include increased length, thickness, pigmentation, and number of lashes or hairs, and misdirected growth of eyelashes. Eyelash changes are reversible upon discontinuation of treatment.

Herpetic Keratitis

Latanoprost should be used with caution in patients with a history of herpetic keratitis, and should be avoided in cases of active herpes simplex keratitis and in patients with a history of recurrent herpetic keratitis specifically associated with prostaglandin analogues.

Macular Oedema

Macular oedema, including cystoid macular oedema, has been reported during treatment with latanoprost. These reports have mainly occurred in aphakic patients, in pseudophakic patients with a torn posterior lens capsule, or in patients with known risk factors for macular oedema. Latacom should be used with caution in these patients.

INTERACTIONS WITH OTHERS MEDICAMENTS:

No specific drug interaction studies have been performed with Latacom. There have been reports of paradoxical elevations in intraocular pressure following the concomitant ophthalmic administration of two prostaglandin analogues. Therefore, the use of two or more prostaglandins, prostaglandin analogues, or prostaglandin derivatives is not recommended.

There is a potential for additive effects resulting in hypotension and/or marked bradycardia when ophthalmic beta-blockers solution is administered concomitantly with oral calcium channel

blockers, beta-adrenergic blocking agents, antiarrhythmics (including amiodarone), digitalis glycosides, parasympathomimetics, guanethidine.

Potentiated systemic beta blockade (e.g., decreased heart rate, depression) has been reported during combined treatment with CYP2D6 inhibitors (e.g. quinidine, fluoxetine, paroxetine) and timolol.

The effect on intraocular pressure or the known effects of systemic beta-blockade may be potentiated when Latacom is given to patients already receiving an oral beta-adrenergic blocking agent, and the use of two or more topical beta-adrenergic blocking agents is not recommended.

Mydriasis resulting from concomitant use of ophthalmic beta-blockers and adrenaline (epinephrine) has been reported occasionally.

The hypertensive reaction to sudden withdrawal of clonidine can be potentiated when taking beta-blockers.

Beta-blockers may increase the hypoglycaemic effect of anti-diabetic agents. Beta-blockers can mask the signs and symptoms of hypoglycaemia.

USE IN PREGNANCY AND LACTATION:

Use in Pregnancy: There are no adequate data from the use of latanoprost and timolol in pregnant women. The potential risk for humans is unknown. Timolol should not be used during pregnancy unless clearly necessary. Epidemiological studies have not revealed malformative effects but show a risk for intra-uterine growth retardation when beta-blockers are administered by the oral route. In addition, signs and symptoms of beta-blockade (e.g.: bradycardia, hypotension, respiratory distress and hypoglycaemia) have been observed in the neonate when beta-blockers have been administered until delivery. If Latacom is administered until delivery, the neonate should be carefully monitored during the first days of life.

Consequently, Latacom should not be used during pregnancy.

Use in Lactation: Beta-blockers are excreted in breast milk. However, at therapeutic doses of timolol in eye drops, it is not likely that sufficient amounts would be present in breast milk to produce clinical symptoms of beta-blockade in the infant. Latanoprost and its metabolites may pass into breast milk.

Latacom should therefore not be used in women who are breast feeding.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Instillation of eye drops may cause transient blurring of vision. Until this has resolved, patients should not drive or use machines.

SIDE EFFECTS:

For latanoprost, the majority of adverse events relate to the ocular system. For timolol, the most serious adverse events are systemic in nature, including bradycardia, arrhythmia, congestive heart failure, bronchospasm and allergic reactions.

Nervous system disorders: Uncommon: Headache

Eye disorders: Very common: Iris pigmentation, Common: Eye irritation (includes stinging, burning, itching, foreign body sensation), eye pain, Uncommon: Eye hyperaemia, conjunctivitis, vision blurred, lacrimation increased, blepharitis, corneal disorders

Skin and subcutaneous tissue disorders: Uncommon: Rash, pruritus

Additional adverse events have been reported specific to the use of the individual components of the medicinal product.

For latanoprost, these are:

Infections and infestations: Herpetic keratitis

Nervous System Disorders: Dizziness

Eye Disorders: Eyelash and vellus hair changes of the eyelid (increased length, thickness, pigmentation, and number of eyelashes), punctate keratitis, periorbital oedema, iritis, uveitis, macular oedema including cystoid macular oedema, dry eye, keratitis, corneal oedema, corneal erosions, trichiasis, iris cyst, photophobia, periorbital and lid changes resulting in deepening of the eyelid sulcus, eyelid oedema, localised skin reaction on the eyelids, pseudophthalmion of the ocular conjunctiva*, darkening of the palpebral skin

Cardiac Disorders: Angina, angina unstable, palpitations

Respiratory, Thoracic and Mediastinal Disorders: Asthma, asthma aggravation, dyspnoea

Musculoskeletal and Connective Tissue Disorders: myalgia/arthritis

General disorders and Administration Site Conditions: Chest pain.

* May be potentially related to the preservative benzalkonium chloride

For timolol, these are:

Immune System Disorders: Systemic allergic reactions including anaphylactic reaction, angioedema, urticaria, localised and generalised rash, pruritus

Metabolism and nutrition disorders: Hypoglycaemia.

Psychiatric Disorders: Insomnia, depression, nightmares, memory loss.

Nervous System Disorders: Syncope, cerebrovascular accident, cerebral ischemia, increases in signs and symptoms of myasthenia gravis, dizziness, paraesthesia, and headache.

Eye Disorders: Signs and symptoms of ocular irritation (e.g. burning, stinging, itching, tearing, redness), blepharitis, keratitis, blurred vision, choroidal detachment following filtration surgery, decreased corneal sensitivity, dry eyes, corneal erosion, ptosis, diplopia.

Ear and Labyrinth Disorders: Tinnitus.

Cardiac Disorders: Bradycardia, chest pain, palpitations, oedema, arrhythmia, congestive heart failure, atrioventricular block, cardiac arrest, cardiac failure.

Vascular Disorders: Hypotension, Raynaud's phenomenon, cold hands and feet.

Respiratory, Thoracic and Mediastinal Disorders: Bronchospasm (predominantly in patients with pre-existing bronchospastic disease), dyspnoea, cough.

Gastrointestinal Disorders: Dysgeusia, nausea, dyspepsia, diarrhoea, dry mouth, abdominal pain, vomiting.

Skin and Subcutaneous Tissue Disorders: Alopecia, psoriasisiform rash, exacerbation of psoriasis, skin rash.

Musculoskeletal and Connective Tissue Disorders: Myalgia

Reproductive System and Breast Disorders: Sexual dysfunction, decreased libido.

General Disorders and Administration Site Conditions: Asthenia, fatigue.

Cases of corneal calcification have been reported very rarely in association with the use of phosphate containing eye drops in some patients with significantly damaged corneas.

SYMPTOMS AND TREATMENT OF OVERDOSAGE:

No data are available in humans with regard to overdose with Latacom.

Symptoms of systemic timolol overdose are: bradycardia, hypotension, bronchospasm and cardiac arrest. If such symptoms occur, the treatment should be symptomatic and supportive. Studies have shown that timolol does not dialyse readily.

Apart from ocular irritation and conjunctival hyperaemia, no other ocular or systemic side effects are known if latanoprost is overdosed. If latanoprost is accidentally ingested orally the following information may be useful:

Treatment: Gastric lavage if needed. Symptomatic treatment. Latanoprost is extensively metabolised during the first pass through the liver. Intravenous infusion of 3 micrograms/kg in healthy volunteers induced no symptoms but a dose of 5.5-10 micrograms/kg caused nausea, abdominal pain, dizziness, fatigue, hot flushes and sweating. These events were mild to moderate in severity and resolved without treatment, within 4 hours after terminating the infusion.

INCOMPATIBILITIES:

Precipitation occurs when eye drops containing thiomersal are mixed with latanoprost. If such drugs are used concomitantly with Latacom, the eye drops should be administered with an interval of at least 5 minutes.

STORAGE CONDITIONS:

Store unopened bottle under refrigeration at 2°C to 8°C. When a bottle is opened for use, it may be stored at room temperature up to 25°C for 4 weeks. Protect from light.

PACK SIZES:

Pack 1's bottle of 2.5 mL in a 5ml eye drop bottle/box.

SHELF LIFE:

Shelf life for unopened bottle: 2 years.

Shelf-life after opening container: 4 weeks

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD

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Taman Klang Jaya, 41200 Klang, Selangor Darul Ehsan, Malaysia